

# Safiri Ai

## AI Internship – Take-Home Assignment (3 Days)

### Predicting Port Congestion and Shipment Delays Using AI

#### Overview

Port congestion is one of the most significant sources of delay and uncertainty in global logistics. Congestion can arise due to a combination of factors including vessel traffic, port capacity, weather conditions, labor constraints, and operational inefficiencies.

These delays have cascading effects across the supply chain, impacting delivery timelines, increasing costs, and reducing reliability for businesses and customers.

At Safiri Ai, we aim to build systems that can anticipate such disruptions and support better planning and decision-making. This assignment evaluates your ability to design an AI-driven approach to understanding and predicting congestion-related delays.

#### Problem Statement

You are given a dataset containing information about vessel movements, port activity, or shipment timelines. Your task is to design and implement a system that can estimate the likelihood of congestion or delay for a given shipment or port.

Your system should:

1. Predict the likelihood of congestion or delay,
2. Provide a confidence score,
3. Explain the key factors contributing to the prediction.

The goal is not only to make predictions, but to make them interpretable and actionable.

#### Recommended Public Datasets

You may use publicly available datasets or create your own synthetic dataset.

One useful dataset source is AIS (Automatic Identification System) vessel tracking data:

<https://marinecadastre.gov/ais/>

AIS data provides information about vessel locations, movement patterns, and port activity, which can be used to derive congestion indicators.

You may also explore port call datasets or shipping-related datasets available on platforms such as Kaggle:

<https://www.kaggle.com/datasets>

If accessing or processing real AIS data is complex, you may construct a synthetic dataset that includes:

- vessel arrival times,
- port capacity constraints,
- queue lengths,
- weather conditions,
- historical delay patterns.

A dataset of 100–500 records is sufficient for this assignment.

## **What You Are Expected to Build**

You should begin by defining what “congestion” means in your context. This could be based on delay thresholds, queue lengths, or deviations from expected turnaround time.

You should then identify relevant features that influence congestion. These may include vessel arrival density, port capacity, historical throughput, or temporal patterns such as time of day or seasonality.

Your system should implement a predictive or analytical approach that estimates the likelihood of congestion or delay. This may involve statistical methods, machine learning models, or rule-based logic, depending on your design.

In addition to prediction, your system should provide explanations. For example, a useful output might indicate that a high number of incoming vessels combined with limited port capacity is likely to result in delays.

## **Suggested Challenges to Address**

You may consider addressing challenges such as:

- temporal dynamics and seasonality,
- incomplete or noisy data,
- variability across different ports,
- interaction between multiple factors such as weather and traffic.

You are not expected to solve all of these, but demonstrating awareness of real-world complexity is valuable.

## **Evaluation**

You should define how your system is evaluated. This may include prediction accuracy, error metrics, or classification performance depending on your formulation.

If you use synthetic data, clearly describe how delays or congestion were simulated and how ground truth was defined.

## **Deliverables**

Your submission should include:

- a code repository with clear instructions,
- a technical report (2–3 pages) describing your approach, assumptions, and results,
- an optional demo (maximum 5 minutes).

## **What We Will Evaluate**

We will evaluate your submission based on your understanding of the problem, the relevance of your features, the soundness of your approach, and your ability to explain your predictions clearly.

We are particularly interested in how you connect data to real-world operational decisions.

## **Constraints**

You should complete this assignment within three days. The solution should be focused, realistic, and clearly justified. Any assumptions made should be explicitly stated.

## **Submission Instructions**

Please submit:

- GitHub repository link,
- PDF report,
- optional demo.

Submission email: `info@safiri.ai`

## **Closing Note**

Port congestion is a complex, multi-factor problem that sits at the intersection of data, operations, and decision-making. At Safiri Ai, we are interested in systems that can simplify this complexity and provide actionable insights.

We are looking for candidates who can think systematically, design practical solutions, and communicate their reasoning effectively.